AI-Based Emergency Call and SMS Triage System for Sri Lankan Disasters

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Emergency call centers often operate on a first-come-first-serve basis, which does not necessarily reflect the true urgency of incidents. As a result, critical cases may remain unattended while less urgent calls are being handled, leading to potentially dangerous delays. This challenge is further intensified in multilingual, low-resource environments such as Sri Lanka, where callers may communicate in Sinhala, Tamil, English, or mixed speech under stressful and noisy conditions.

This research proposes a **Multimodal Urgency Detection Module using Prosodic and Conversational Cues for Emergency Call Prioritization**, designed to support rapid and accurate urgency assessment. The system operates in a near-real-time setting, where callers first interact with an AI agent, and the conversation is either analysed incrementally during the call or recorded and summarized for immediate analysis.

The proposed approach integrates prosodic speech features, such as pitch, intensity, jitter, shimmer, and speech rate, with rule-based conversational cues extracted from summaries or transcripts. These multimodal features are fused and used to classify calls into High, Medium, and Low urgency levels. The results are presented on an operator dashboard, enabling efficient prioritization of queued calls.

The system will be evaluated using a simulated multilingual dataset, with potential extension to real-world data. The expected outcomes include improved prioritization accuracy, reduced operator workload, and faster response to critical situations in low-resource emergency environments.

Keywords: Emergency calls, urgency detection, multimodal analysis, prosodic features, conversational cues, queue prioritization, multilingual speech processing, Sinhala, Tamil, low-resource languages

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence
AIMS – Autonomous Intelligent Machines and Systems
API – Application Programming Interface
ASR – Automatic Speech Recognition
DMC – Disaster Management Centre
FCFS – First-Come-First-Serve
GeMAPS – Geneva Minimalistic Acoustic Parameter Set
IP – Intellectual Property
LKR – Sri Lankan Rupee
ML – Machine Learning
MLP – Multilayer Perceptron
NDMA – National Disaster Management Authority
NDRF – National Disaster Response Force
NIPO – National Intellectual Property Office
NLP – Natural Language Processing
RF – Random Forest
SaaS – Software as a Service
SDG – Sustainable Development Goal
SLIIT – Sri Lanka Institute of Information Technology
SMS – Short Message Service
SVM – Support Vector Machine
WBS – Work Breakdown Structure

1. INTRODUCTION

1.1 Background and context

Floods, landslides, cyclones, medical emergencies, and accidents contribute to a rising number of emergency hotline calls in countries such as Sri Lanka. During such events, call centers experience sudden surges in demand, requiring operators to quickly gather critical information such as location, incident type, number of affected individuals, and level of danger. However, callers are often under panic, emotional distress, or physical threat, resulting in unclear or incomplete communication, which delays emergency response.

This challenge is further intensified by Sri Lanka's multilingual environment, where callers may use Sinhala, Tamil, English, or code-mixed forms. Under stress, they may switch languages or provide fragmented information, while background noise and network instability further hinder communication. As a result, operators often need to ask follow-up questions, delaying time-sensitive responses.

In addition, most emergency call centers operate on a **first-come-first-serve basis**, which does not reflect the true urgency of incidents. As a result, operators may handle lower-priority calls while more critical cases remain in the queue, leading to serious consequences. Current systems rely heavily on human operators to interpret information and assess urgency, increasing workload and reducing efficiency, particularly during peak demand.

Existing technological solutions are mainly limited to speech transcription or basic call routing and lack intelligent urgency assessment capabilities. In practice, urgency is conveyed through both **what the caller says** and **how the caller speaks**, and analyzing these signals independently can lead to inaccurate prioritization.

To address these challenges, this research proposes a **Multimodal Urgency Detection Module using Prosodic and Conversational Cues for Emergency Call Prioritization**, developed as part of a broader AI-Based Emergency Call and SMS Triage System. The module operates in a **near-real-time setting**, where callers first interact with an AI agent and the conversation is analyzed incrementally where possible, with rapid post-call assessment using recorded audio and AI-generated summaries. Prosodic features and rule-based conversational cues are combined through multimodal fusion to classify calls into High, Medium, and Low urgency levels.

The results are presented on an operator dashboard organized by urgency, enabling faster identification of critical cases and supporting informed callback and response decisions. This research focuses on Sri Lanka's multilingual, low-resource context and contributes to improving emergency response through AI-driven decision support, aligning with **SDG 3, SDG 9, SDG 11**, and the **AIMS research cluster**.

1.2 Literature Review

Several research studies have explored the use of artificial intelligence, speech processing, and natural language processing techniques to improve emergency call analysis and triage systems.

Speech-based analysis, particularly prosodic feature extraction, has been widely used to detect emotional stress in emergency calls. [1] demonstrated that acoustic features such as pitch, intensity, and voice stability vary significantly under stress in real emergency conversations, indicating that prosodic signals can provide useful cues about the caller's condition. Similarly, [2] showed that speech emotion recognition systems can effectively utilize acoustic features to identify emotional states such as distress, fear, and urgency.

Emotion recognition in emergency call environments has also been investigated. Deschamps-[3] explored multimodal emotion recognition in emergency call center data and highlighted the challenges posed by noisy environments and spontaneous speech. Their findings suggest that combining multiple modalities can improve performance compared to single-modality systems.

In addition to speech-based approaches, text-based analysis has been used to extract meaningful information from emergency conversations [4] introduced Whisper, a robust multilingual speech recognition system capable of generating accurate transcriptions even in noisy conditions, making it suitable for emergency call scenarios. Textual analysis techniques can then be applied to these transcriptions to identify important conversational cues related to urgency.

Multimodal approaches that combine speech and text features have shown improved performance in classification tasks. [5] demonstrated that integrating audio and textual information leads to better sentiment and emotion classification results compared to unimodal approaches. Similarly, [6] introduced openSMILE, a widely used toolkit for extracting acoustic features such as those defined in the GeMAPS feature set, which are commonly applied in speech emotion and stress analysis.

Despite these advancements, most existing approaches rely heavily on data-driven machine learning or deep learning models, which require large annotated datasets. This presents a challenge for low-resource languages such as Sinhala and Tamil, where such datasets are limited. As noted by [7], Sinhala remains a low-resource language in NLP, and similar limitations exist for Tamil, making it difficult to directly apply complex data-intensive models.

Overall, prior work demonstrates significant progress in speech emotion recognition, text-based analysis, and multimodal feature fusion for emergency call processing. However, there is limited research on integrating prosodic speech features with rule-based conversational cue detection for real-time urgency classification in low-resource multilingual environments.

1.3 Research Gap

Despite the advancements in emergency call analysis systems, several key limitations remain.

Firstly, many existing systems focus on either speech-based analysis or text-based analysis independently. While speech emotion recognition systems analyze prosodic features and text-based systems analyze conversational content, few approaches effectively combine both modalities for urgency classification. However, accurately determining urgency requires integrating both how the caller speaks and what the caller says [1] [5] [6].

Secondly, most existing approaches rely on deep learning models that require large annotated datasets. This is a significant limitation for low-resource languages such as Sinhala and Tamil, where labeled emergency call data is scarce [7]. As a result, such models may not perform reliably in real-world multilingual emergency scenarios.

Thirdly, emotion detection alone is not sufficient for determining urgency. A caller may sound calm during a life-threatening situation or highly emotional in a less critical case. Therefore, relying solely on emotional cues can lead to incorrect prioritization [2] [3]

Fourthly, existing systems are not designed to support queue-aware prioritization in emergency call centers. Most current workflows operate on a first-come-first-serve basis and do not incorporate automated urgency assessment to assist operators in identifying high-priority cases waiting in the queue. During large-scale emergencies, call volumes can exceed operator capacity, causing delays in service and highlighting the need for priority-based queue management [8].

Finally, most research focuses on complex machine learning architectures, with limited exploration of lightweight and interpretable approaches. In emergency contexts, rule-based detection of critical conversational cues (e.g., “fire,” “accident,” “trapped,” “drowning,” “bleeding”) can provide a practical and efficient alternative, especially when combined with prosodic speech features.

Therefore, there is a clear need for a **multimodal urgency detection system** that integrates prosodic speech analysis with rule-based conversational cue detection for Sinhala and Tamil emergency calls. The proposed system addresses this gap by providing a **real-time or near real-time, low-resource, and interpretable solution** for classifying emergency calls into urgency levels to support operator decision-making.

2. OBJECTIVE

2.1 Main Objective

To develop a **Multimodal Urgency Detection Module** capable of automatically classifying emergency calls into **High**, **Medium** and **Low** urgency levels by analyzing **prosodic features** and **conversational cues** in Sinhala and Tamil, enabling faster and more accurate emergency triage in Sri Lankan low-resource language environments.

2.2 Specific Objectives

- To extract prosodic features such as pitch, intensity, jitter, shimmer, and speech rate from caller audio and analyze their relationship to urgency levels using openSMILE and GeMAPS.
- To obtain conversational cues from Whisper speech-to-text outputs and apply multilingual NLP techniques to detect urgency-related keywords, semantic patterns and contextual signals in Sinhala and Tamil.
- To design and develop a **hybrid fusion model** that integrates prosodic and linguistic features to generate a combined urgency score in real time.
- To construct and evaluate a simulated multilingual emergency call dataset, and incorporate real call data if available through authorized channels.
- To measure the model's performance using accuracy, precision, recall, F1-score and false negative rate, with a specific emphasis on minimizing misclassification of high-urgency calls.
- To integrate the urgency detection module into the larger emergency triage workflow and ensure compatibility with downstream components of the proposed AI-based emergency communication system.

3. METHODOLOGY

3.1 Research Approach

In this study, the objective is to develop a **Multimodal Urgency Detection Module** to support emergency call triage. The research is categorized as **applied and experimental research**, as it aims to build a functional system and evaluate its effectiveness in classifying emergency calls based on urgency levels. The proposed system integrates both **speech processing** and **natural language processing techniques** to analyze emergency call data.

A **multimodal approach** is employed, combining two key sources of information: (i) **prosodic speech features**, which capture how the caller speaks, and (ii) **conversational cues**, which capture what the caller says.

Given the low-resource nature of Sinhala and Tamil languages, this research adopts a **rule-based approach** for conversational cue detection instead of relying on data-intensive deep learning models. This ensures that the system remains **efficient, interpretable, and practical for real-world deployment**, especially in environments where labeled datasets are limited.

The system is designed to operate in **real-time or near real-time**, enabling urgency classification either during the call or immediately after it is recorded. The effectiveness of the system is evaluated using multiple machine learning models and standard evaluation metrics, with particular emphasis on minimizing errors in high-urgency cases.

3.2 Technical Methods and Algorithms

The proposed system uses a **hybrid multimodal pipeline**, integrating prosodic features from raw audio and conversational cues from speech transcripts. The pipeline consists of the following stages:



Figure 1-Proposed Multimodal Urgency Detection Pipeline

1. **Audio Preprocessing** -The emergency call audio is first subjected to preprocessing to improve quality and ensure consistency. This includes **noise reduction**, **amplitude normalization**, and **segmentation into short frames** suitable for feature extraction. These steps are essential for handling real-world emergency conditions, such as background noise, varying audio quality, and overlapping speech.
2. **Speech-to-Text Transcription** - The preprocessed audio is converted into text using the **Whisper Automatic Speech Recognition (ASR)** model. Whisper is selected due to its ability to handle **multilingual speech**, including Sinhala and Tamil, and its robustness in noisy environments. The generated transcripts serve as input for conversational analysis.
3. **Prosodic Feature Extraction** - Prosodic features related to stress and urgency are extracted from the audio using the **openSMILE toolkit** in combination with the **GeMAPS feature set**. These features include pitch, intensity, jitter, shimmer, speech rate, and harmonics-to-noise ratio. Such features are widely used in speech analysis research and have been shown to correlate with emotional stress and urgency in emergency scenarios. They provide critical information about the caller's vocal behavior, which is an important indicator of urgency.
4. **Conversational Cue Detection** -The transcribed text is analyzed using a **rule-based approach** to identify urgency-related linguistic cues. This includes detecting:

• Emergency-related keywords (e.g., *fire, accident, drowning, trapped, bleeding*)

• Severity indicators (e.g., *not breathing, unconscious, urgent help*)

• Simple pattern-based expressions indicating critical situations

This approach is particularly suitable for **low-resource languages**, as it does not require large annotated datasets. It also provides **interpretability**, allowing the system to clearly justify why a call is classified as urgent.

5. **Multimodal Feature Fusion** - The extracted prosodic features and conversational cues are combined using a **feature-level fusion approach**. This involves concatenating the feature vectors from both modalities into a single unified representation. Before fusion, features are normalized to ensure compatibility across different scales. This multimodal representation enables the system to leverage both acoustic and linguistic information for improved classification performance.

6. **Urgency Classification**- The fused feature set is used as input to machine learning models to classify emergency calls into three categories:

• **High urgency**

• **Medium urgency**

• **Low urgency**

Multiple models are evaluated, including **Random Forest**, **Support Vector Machine (SVM)**, and **Neural Networks (MLP)**, to determine the most effective approach.

The classification system is optimized to minimize **false negatives for high-urgency cases**, ensuring that critical situations are not overlooked. The system is designed to provide results in **real-time or near real-time**, supporting rapid decision-making in emergency call centers.

3.3 Tools and Platforms with Justification

Technology	Purpose	Justification
Whisper ASR	Speech transcription	Provides robust multilingual transcription and performs well in noisy emergency environments
openSMILE	Acoustic feature extraction	Industry-standard toolkit in speech emotion and stress analysis, widely used in prosody research [1]
GeMAPS	Prosodic feature set	Designed specifically for affective speech analysis, capturing key stress and urgency indicators
ML Models (RF, SVM, Neural Networks)	Urgency classification	Enables comparison of classical and deep learning models for optimal performance
PyTorch / TensorFlow	Model training and fusion learning	Supports deep learning architectures required for multimodal fusion
FastAPI / Flask	API deployment	Provides scalable integration with the larger emergency triage system
PostgreSQL / MongoDB	Data management	Secure storage for audio, transcripts and feature vectors

Table 1-Tools and Platforms

3.4 Data Collection and Ethical Considerations

- **Data Sources:**
    - Because real emergency call data is sensitive and not immediately available, the project will use:
- **Simulated Emergency Calls**
    - Recorded by volunteers in Sinhala and Tamil
    - Multiple dialects, genders, ages
    - Various acoustic conditions (quiet, noisy, outdoor, phone-quality)
    - Realistic emergency scenarios (accidents, floods, medical distress)
- **Synthetic Data Augmentation**
    - Noise injection (sirens, wind, traffic)
    - Speed/pitch alteration
    - Room impulse simulation
- **Potential Real DMC Dataset**
    - Access request procedure is underway
    - If approved, real calls will be anonymized and integrated for evaluation

- **Ethical considerations**
- Informed consent from all volunteer participants
- No personally identifiable information stored
- Audio anonymization for real caller data
- Secure encrypted storage
- Compliance with SLIIT Research Ethics Committee guidelines
- No system decision overrides human operator judgment

3.5 Validation Strategy and Performance Metrics

To evaluate the effectiveness of the urgency detection module, the following strategies and metrics will be used:

1. Quantitative Evaluation

- **Accuracy** — overall correctness of urgency predictions
- **Precision, Recall, F1-score** — especially for *High Urgency* class
- **False Negative Rate** — critical in emergency settings
- **Confusion Matrix Analysis** — to assess misclassifications

2. Qualitative Evaluation

- Human operator feedback on urgency labels
- Evaluation during simulated emergency-call sessions
- Consistency analysis across Sinhala and Tamil dialects

3. Cross-Validation

- Stratified K-fold validation on simulated dataset
- If real data becomes available, evaluate on matched distributions

4. Benchmarking

- Compare performance of:
    - Audio-only model
    - Text-only model
    - Multimodal fusion model

Prior multimodal emergency-call research shows that fused models outperform single-modality systems

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed **Multimodal Urgency Detection Module** is designed as a core component of an AI-based emergency communication system. It operates as a structured pipeline that integrates both **speech-based** and **text-based analysis** to automatically determine the urgency level of an incoming emergency call. The overall workflow is illustrated in the methodology diagram.

The process begins when an emergency call is received through the system and captured as raw audio. This audio may contain background noise, overlapping speech, and variations in speaking style due to stress. The system processes this input through two complementary analytical streams: **prosodic acoustic analysis** and **conversational cue analysis**.

The first stage involves **speech-to-text transcription**, performed using the Whisper Automatic Speech Recognition (ASR) model. Whisper converts multilingual speech, including Sinhala, Tamil, and code-mixed inputs, into text while maintaining robustness in noisy environments. This ensures reliable transcription even when callers speak rapidly or under distress.

In parallel, the system performs **prosodic feature extraction** using openSMILE in conjunction with the GeMAPS feature set. This stage extracts acoustic features such as pitch, jitter, shimmer, loudness, and speech rate. These features capture variations in vocal patterns that are commonly associated with stress and urgency in emergency situations.

The generated transcripts are then processed by a **Conversational Cue Detection module**, which applies a **rule-based approach** to identify urgency-related linguistic signals. This includes detecting emergency keywords (e.g., *fire, accident, drowning, trapped*), severity indicators (e.g., *not breathing, unconscious*), and simple pattern-based expressions. This approach is particularly suitable for low-resource languages, as it does not depend on large annotated datasets and provides interpretable results.

Following feature extraction, both prosodic and conversational features are combined using a **Multimodal Feature Fusion layer**. This fusion integrates acoustic and linguistic information into a unified representation, enabling the system to analyze both how the caller speaks and what the caller says simultaneously.

The fused feature vector is then passed to a **Machine Learning Classification module**, which categorizes the emergency call into one of three urgency levels: **High**, **Medium**, or **Low**. Multiple models, including Random Forest, Support Vector Machine (SVM), and Neural Networks, are evaluated to determine the best-performing approach. The system is specifically optimized to achieve high recall for high-urgency cases, reducing the risk of critical misclassification.

The entire system is designed to operate in **real-time or near real-time**, providing rapid urgency predictions to support emergency operators in decision-making. By integrating prosodic speech analysis with rule-based conversational cue detection, the proposed module

offers a **lightweight, interpretable, and effective solution** for emergency call triage in multilingual, low-resource environments such as Sri Lanka.

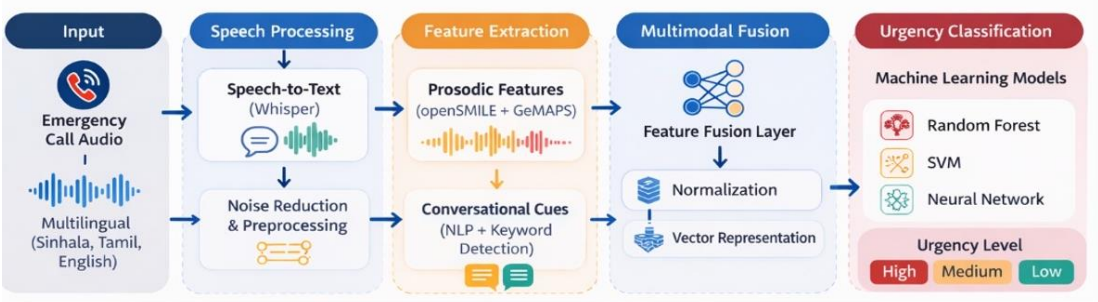


Figure 2-System Architecture

5. USER REQUIREMENTS

5.1 Stakeholder Identification

1. **Emergency Call Operators**
   Individuals who receive incoming emergency calls and rely on system-generated urgency levels to prioritize responses.
2. **Disaster Management Centre (DMC) Supervisors**
   Authorities who oversee emergency operations and require accurate urgency insights for strategic decision-making.
3. **Civilians / Callers**
   Indirect beneficiaries whose calls are analyzed by the system to ensure quick and efficient emergency response.
4. **First Responders**
   Medical teams, firefighters and rescue personnel who depend on accurate urgency classification to respond appropriately.
5. **System Administrators / Developers**
   Technical personnel responsible for maintaining, updating and integrating the urgency detection module into the full triage system.

5.2 Functional Requirements

1. **Multilingual Audio Processing**
   The system must process Sinhala, Tamil, English and mixed-language emergency calls.
2. **Speech-to-Text Conversion**
   The system must transcribe emergency call audio into text using Whisper ASR.
3. **Prosodic Feature Extraction**
   The system must extract acoustic features such as pitch, jitter, shimmer, intensity and speech rate.
4. **Conversational Cue Extraction**
   The system must detect linguistic urgency indicators from transcribed speech using multilingual NLP.
5. **Multimodal Feature Fusion**
   The system must combine prosodic and linguistic features to form a unified urgency representation.
6. **Urgency Classification**
   The system must categorize emergency calls as High, Medium or Low urgency.
7. **Real-Time Operation**
   The system must deliver results within a few seconds of receiving the audio input.

8. **Robust Noise Handling**
   The system must operate effectively under noisy, chaotic or low-quality audio conditions.
9. **API-Based Integration**
   The system must expose prediction results through secure APIs for use by other system components.
10. **Complete Data Logging**
    The system must store audio, transcripts, extracted features and outputs for auditing and model evaluation

5.3 Non-Functional Requirements

1. **Performance** -The system should deliver urgency predictions rapidly (within 2–4 seconds).
2. **Accuracy**- The system should maintain high accuracy across all urgency levels, with special emphasis on minimizing false negatives for High Urgency calls.
3. **Reliability**- The system should function consistently during peak call loads and stressed operational conditions.
4. **Scalability**- The system should support multiple simultaneous call analyses without performance degradation.
5. **Security**- All transmitted and stored data must be encrypted and anonymized where necessary.
6. **Usability** -The system's output should be intuitive, clear and easily interpretable by emergency operators.
7. **Resilience**- The system should provide stable performance even with incomplete sentences, mixed languages or partially corrupted audio.
8. **Ethical Compliance** -The system should adhere to SLIIT and DMC ethical guidelines and must not override human decision-making.

6. COMMERCIALIZATION PLAN

The major customer is the Disaster Management Centre (DMC), which is a government agency with the legal responsibility to respond to disasters. The major decision-makers are the Director General of the DMC and the IT and Operations divisions. The system addresses the business need of the DMC to speed up the response time and information overload in the response to disasters. There is also **a secondary market** of provincial emergency operations centers in the country's 9 provinces. There is also an **export market** consisting of disaster response agencies in the countries of South Asia, including the Bangladesh NDMA and the India-based National Disaster Response Force (NDRF), which also face the challenge of dealing with multilingual communications in low-resource languages.

The product **offers three value propositions** that can be quantified. These are: Speed – reducing the time taken to process duplicate reports from 15-20 minutes to less than 8 seconds per event.

Safety – ensuring that critical events are not missed due to information overload with the urgency-sensitive safety architecture.

Intelligence – enabling DMC operators to make decisions to dispatch resources with the structured intelligence from the incidents instead of raw call records.

The revenue streams align with the Government Technology (GovTech) Software as a Service Licensing Model, which is suitable for the Sri Lanka Public Sector Procurement Environment. The revenue streams are tiered as follows:

(1) Years 1-2: Research Pilot with DMC under a sponsored research agreement, with no license fee, funded by NSF Sri Lanka;

(2) Years 3-4: national deployment license with DMC at 3.5M LKR per year, including 1 year of hosting, support, and updates; and

(3) Years 5+: regional export license with Bangladesh NDMA and India NDRF at 50,000 - 120,000 USD per year per country.

The data insights subscription service offers an additional revenue stream of 1.2M LKR per year at the national level.

The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support. In the national deployment phase (Years 3 to 4), the system is planned to be offered as an annual Software as a Service (SaaS) license to DMC and Provincial EOCs in Sri Lanka, priced annually at an estimated LKR 3.5 million, which also includes system maintenance and infrastructure

services. In the regional expansion phase (Year 5 and onward), the system is planned to be adapted and offered to international disaster management organizations such as the Bangladesh National Disaster Management Authority and India's National Disaster Response Force, priced annually between $50,000 and $120,000 as a regional license depending on the scale of deployment and customization required. In addition to the above services offered to DMC and other research partners, an Analytics Add-on service is planned to be offered to DMC and research partners for post-disaster data analysis and insights reporting, priced annually at approximately LKR 1.2 million.

The competitive advantages of the system are:

(1) the first-mover advantage in the multilingual emergency fusion marketplace of South Asia—no competing product speaks to Sinhala or Tamil;

(2) the Sri Lanka emergency location gazetteer is a proprietary data asset that cannot be replicated without years of linguistics fieldwork;

(3) the open-source technology stack of LaBSE, DBSCAN, PostgreSQL, and React minimizes the cost of licenses; (4) the urgency-sensitive safety architecture is a patentable new design that has yet to be replicated by competitors.

Three IP assets to be developed:

(1) Sri Lanka emergency locations gazetteer to be filed as a data publication and issued a suitable Creative Commons license for research use;

(2) urgency-sensitive DBSCAN clustering algorithm to be filed as a provisional patent with the National Intellectual Property Office of Sri Lanka (NIPO) after system completion;

(3) research paper to be published to claim scientific priority. The university IP policy and the sponsored research agreement between DMC and SLIIT are applicable.

7. BUDGET AND JUSTIFICATION

Category	Total (LKR)
Software Licenses	60,000
Cloud Services	30,000
Commercialization & IP	20,000
Miscellaneous	10,000
TOTAL 1-YEAR PROJECT BUDGET	**LKR 120000**

Table 2: BUDGET AND JUSTIFICATION

8. WORK BREAKDOWN STRUCTURE (WBS)

Phase / Timeline	Component	Responsible Member	Key Tasks	Workload Contribution
Month 1 – Month 2	Project Planning	All Members	Literature Review, Requirement Analysis, System Architecture Design, Dataset Planning	Shared
Month 3 – Month 5	Voice-Based Language Identification, Incident Understanding & Urgency Detection	Member 1	Audio Preprocessing, Speech-to-Text Transcription, Multilingual Language Detection, Panic/Prosody Analysis, Incident Classification, Location Information, Urgency Detection, Generating Structured JSON Output	Core responsibility of Member 1
Month 3 – Month 5	Multilingual Emergency AI Clarification Agent	Member 2	AI Conversation Agent Design, Information Not Provided, Question Generation, Multilingual Conversation Support (Sinhala, Tamil, English), Panic-Aware Conversation, Handling Interruptions and Call	Core responsibility of Member 2

			Drops, Generating Structured Conversation Summaries	
Month 3 – Month 5	Low-Bandwidth Text/SMS Emergency Intelligence	Member 3	Language Detection for Short Messages, Normalization of Short Message Text, Handling Emojis and Abbreviations, Handling Romanized Sinhala and Tamil, Incident Information, Urgency, Generating Structured SMS Intelligence Output	Core responsibility of Member 3
Month 6 – Month 7	Multimodal Incident Fusion, Timeline Reconstruction & Dashboard	Member 4	Integration of output from components 1-3, incident clustering, duplicate report detection, conflict resolution, creation of a unified incident record, timeline reconstruction, confidence scoring, visualization using a dashboard	Core responsibility of Member 4
Month 8	System Integration	All Members	Integration of all four components,	Shared

			communication via API, testing data flow between modules	
Month 9	System Testing & Evaluation	All Members	Model evaluation, testing, measuring, validation	Shared
Month 10	Dashboard Improvement	Member 4+ Team	Improvement in visualizations, usability testing, refinement	Shared with lead Member 4
Month 11	System Optimization	All Members	Optimization, bug fixes, testing	Shared
Month 12	Documentation & Presentation	All Members	Writing final report, documentation, research paper, final presentation	Shared

Table 3: WORK BREAKDOWN STRUCTURE (WBS)

9. GANTT CHART

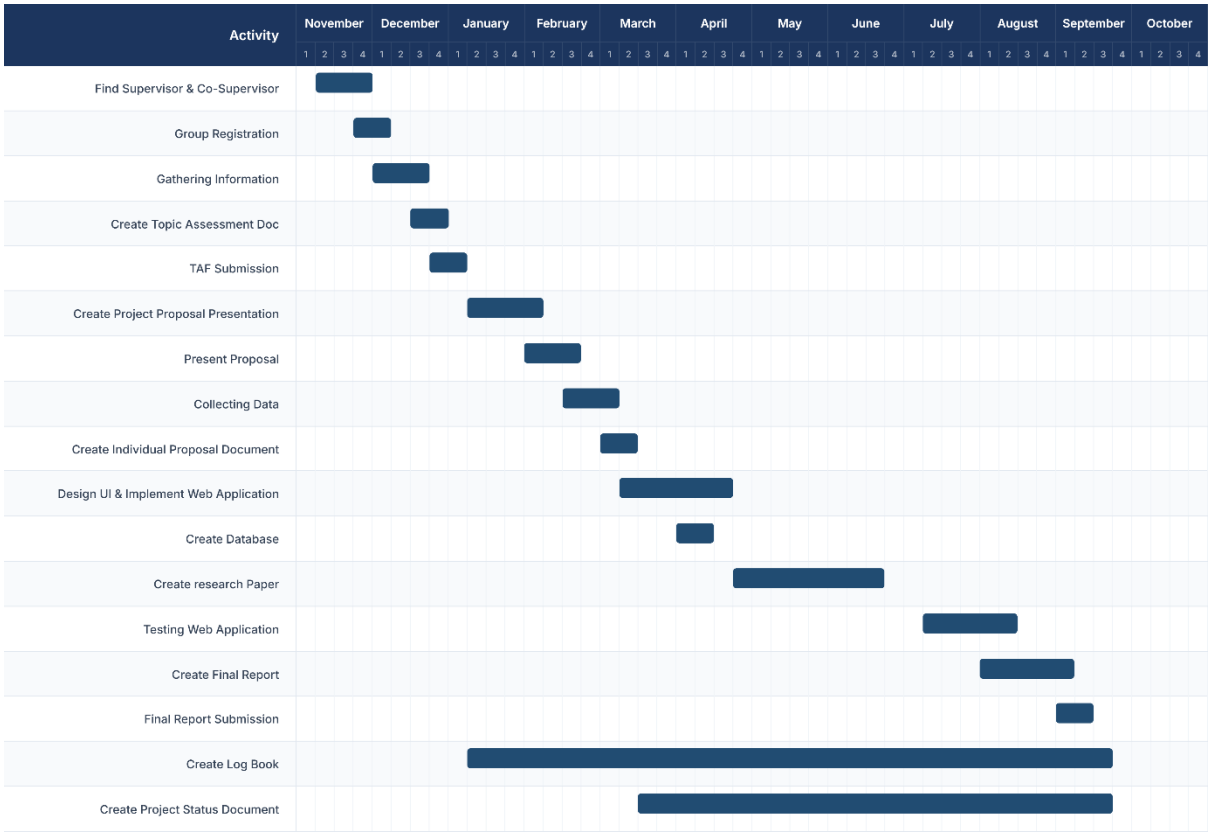


Figure 3: Gantt Chart

10.APPENDICES

10.1 Preliminary UI Mockups

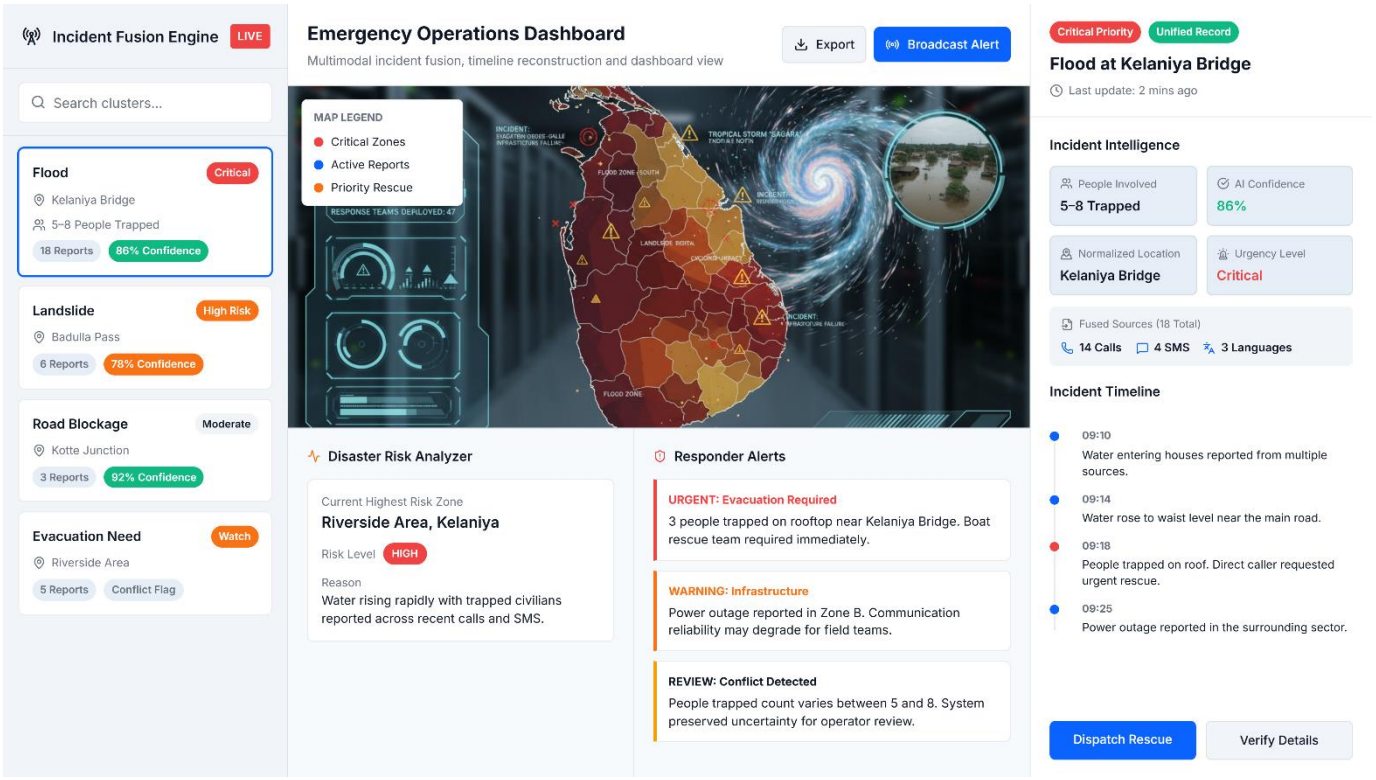


Figure 4: Pre liminary UI Mockups

11.References

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12.RISK ANALYSIS

The importance of risk analysis lies in identifying possible risks that may pose a challenge to the development and operation of the proposed System. By identifying these risks, appropriate mitigation measures can be established to minimize their effect on the system.

1. Technical Risks

The system may struggle with noisy or low-quality emergency call audio, causing inaccuracies in Whisper transcription or prosodic feature extraction. Simulated data may also limit model generalization until real emergency call data becomes available, and multimodal fusion errors could reduce classification accuracy if acoustic and linguistic features do not align.

2. Operational Risks

Real emergency callers may speak unclearly, switch languages frequently, or provide incomplete information due to panic, which can affect urgency detection. Call drops, interrupted speech or rapid delivery may also reduce model reliability, and any misclassification of high-urgency calls poses immediate operational risks.

3. Data Privacy and Security Risks

Emergency call audio contains sensitive personal information that must be securely stored and anonymized. Unauthorized access, improper handling of recordings or inadequate encryption could expose private caller data and violate ethical guidelines.

4. Scalability Risks

During peak disaster periods, the system may receive multiple simultaneous calls, increasing computational load. High resource demands for multimodal analysis may cause processing delays or reduced real-time performance under heavy traffic.

5. Integration Risks

Integration challenges may occur when connecting the urgency detection module with the full emergency triage system, dashboards or APIs. Operator workflows may need adaptation, and version conflicts or system incompatibility could disrupt overall system functionality